$\alpha \quad |\vec{b}| = \sqrt{b_x^2 + b_y^2} = b$

$\alpha \quad \phi = \tan^{-1}\left(\frac{b_y}{b_x}\right)$

$\alpha \quad \vec{r} = \vec{a} + \vec{b} \begin{cases} r_x = a_x + b_x \\ r_y = a_y + b_y \\ r_z = a_z + b_z \end{cases}$

$\alpha \quad \vec{a}.\vec{b} = \vec{b}.\vec{a} = a.b.\cos\phi$ (noktasal skale çarpım)

$\hat{i}.\hat{i} = 1 \quad \hat{i}.\hat{j} = 0$

$\hat{j}.\hat{j} = 1 \quad \hat{i}.\hat{k} = 0$

$\hat{k}.\hat{k} = 1 \quad \hat{k}.\hat{j} = 0$

$[\hat{i}] = [\hat{j}] = [\hat{k}] = 1$

$a = \sqrt{a^2} = \sqrt{\vec{a}.\vec{a}} = \sqrt{a_x^2 + a_y^2 + a_z^2}$

$\vec{a}.\vec{b} = (a_x\hat{i} + a_y\hat{j} + a_z\hat{k}).(b_x\hat{i} + b_y\hat{j} + b_z\hat{k})$

$= a_x.b_x + a_y.b_y + a_z.b_z$

$\frac{\vec{a}.\vec{b}}{a.b} = \cos\phi \rightarrow \arccos\left(\frac{\vec{a}.\vec{b}}{ab}\right) = \phi$

$\alpha \quad \vec{a} \times \vec{b} = \vec{c} \quad c = a.b.\sin\phi$ (hacim vektörel çarpım)

$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

$\vec{c}$ nin yönü $\vec{a}$ ve $\vec{b}$ nin düzlemine DİKTİR.

ϕ (+) saat tersi

$\phi \; \vec{c}.\vec{a} = 0$

$\vec{c}.\vec{b} = 0$

$\hat{i} \times \hat{i} = 0 \quad \hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{j} = 0 \quad \hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{k} = 0 \quad \hat{k} \times \hat{i} = \hat{j}$

$k - \underbrace{i - j}_{\text{sıralama (+)}} - k$

$\hat{i} \times \hat{k} = -\hat{j}$

$\vec{a} \times \vec{b} = (a_x\hat{i} + a_y\hat{j} + a_z\hat{k}) \times (b_x\hat{i} + b_y\hat{j} + b_z\hat{k})$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y)\hat{i} - (a_x b_z - a_z b_x)\hat{j} + (a_x b_y - a_y b_x)\hat{k}$$

∝ Δx = yer değiştirme = $x_s - x_i$

∝ V_{ort} = ortalama hız = $\frac{\Delta x}{\Delta t}$ → aynı yönlü

∝ S_{ort} = ortalama sürat = $\frac{\text{toplam mesafe}}{\Delta t}$

∝ a —(alan, eğim, türev)→ V —(alan, eğim, türev)→ x

∝ V = anlık hız = $\frac{dx}{dt} = x'(t)$

∝ anlık sürat = $|V|$

∝ a_{ort} = ortalama ivme = $\frac{\Delta V}{\Delta t}$ → aynı yönlü

∝ a = anlık ivme = $\frac{dV}{dt} = V'(t)$

∝ Sabit ivmeli hareketin denklemleri:

$$V_s = V_i + at \quad , \quad x_s - x_i = V_i \cdot t + \frac{1}{2} a t^2$$

$$V_s^2 = V_i^2 + 2a(x_s - x_i) \Rightarrow V^2 = V_0^2 + 2a(\Delta x)$$

∝ $V_s = V_0 + at$

∝ $x = V_0 \cdot t$

∝ Sabit düzeme ivmeli:

$$V = V_0 - gt \quad , \quad y - y_0 = V_0 \cdot t - \frac{1}{2} g t^2$$

$$V^2 = V_0^2 - 2g(y - y_0)$$

∝ $\Delta \vec{r} = \vec{r}_s - \vec{r}_i = (x_s - x_i)\hat{i} + (y_s - y_i)\hat{j} + (z_s - z_i)\hat{k}$

— Fırlatma —

yörünge denklemi = $y = \tan\phi - \frac{g}{2(V_0 \cdot \cos\phi)^2} \cdot x^2$

menzil denklemi = $R = \frac{V_0^2}{g} \cdot \sin 2\phi$

$V_y = V_{0y} - gt$

---

düzgün dairesel

∝ $a_{mr} = \frac{V^2}{r} = \omega^2 \cdot r \Rightarrow \frac{4\pi^2 \cdot r}{T^2}$ $\quad \propto V = \omega \cdot r \quad \omega = \frac{2\pi}{T}$

∝ $T = \frac{2\pi r}{V}$ $\quad \propto F_{net} = m \cdot a = m \cdot \frac{V^2}{r}$

---

bağıl hareket

∝ $V_{gözlen} - V_{gözlemci}$ $\quad \propto V_{BA}$ (nen ← y ci)

∝ $X_{PA} = X_{PB} + X_{BA}$, $V_{PA} = V_{PB} + V_{BA}$, $a_{PA} = a_{PB}$

Kuvvet Hor.

$\vec{F}_{net} = m\vec{a}$

SI(MKS)	m	kg	m/s^2	Newton $= 1\ kg.m/s^2$
SI(CGS)	cm	g	cm/s^2	dyne $= 1\ gr.cm/s^2$

$1N = 10^5$ dyne

$F_{net} = 0$ } I. Kanun

(Cisim durmakta, durmaya devam eder, gidiyorsa gitmeye devam eder (sabit hizla)

$F_{net} = m.a$ } II. Kanun

$\vec{F} = m\vec{a}$

Cisimler etkilesiyorsa, cisimler birbirine ayni buyuklukte fakat ters yonlu kuvvet uygular. } III. Kanun etki - tepki

etki - tepki tek bir cisme etki etmez (normal kuvvet - kutle cekim etki - tepki degildir)

$f_s = \mu N$

2024

1) $\vec{A} = -2\hat{i} + 3\hat{j} - 3\hat{k}$, $\vec{B} = 3\hat{i} + 2\hat{j} - 2\hat{k}$ ve $\vec{C} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

a) $\vec{R_1} = \vec{A} + 2\vec{B} - \vec{C} = ?$

$-2\hat{i} + 3\hat{j} - 3\hat{k} + 6\hat{i} + 4\hat{j} - 4\hat{k} - 2\hat{i} - 3\hat{j} - 4\hat{k}$

$\vec{R_1} = 2\hat{i} + 4\hat{j} - 11\hat{k}$ /1

b) vektörlerin büyüklüğü?

$|\vec{A}| = \sqrt{(-2)^2 + 3^2 + (-3)^2} = \sqrt{4+9+9} = \sqrt{22} = 4,6904 = 4,69 = A$ /1

$|\vec{B}| = \sqrt{3^2 + 2^2 + (-2)^2} = 4,123 = 4,12 = B$ /1

$|\vec{C}| = \sqrt{2^2 + 3^2 + 4^2} = 5,385 = 5,38 = C$ /1

c) $R_2 = \vec{A} \cdot \vec{B} = ?$ (skalar çarpım)

$R_2 = (-2\hat{i} + 3\hat{j} - 3\hat{k}) \cdot (\vec{B} = 3\hat{i} + 2\hat{j} - 2\hat{k})$

$R_2 = (-6)\hat{i}\hat{i} + 6\hat{j}\hat{j} + 6\hat{k}\hat{k} = -6 + 6 + 6 = 6$ $\quad R_2 = 6$ /1

(1 ? 1 ? 1)

d) A ve B vektörleri arasındaki açı?

$\vec{A} = -2\hat{i} + 3\hat{j} - 3\hat{k}$, $\vec{B} = 3\hat{i} + 2\hat{j} - 2\hat{k}$

$\vec{A} \cdot \vec{B} = AB\cos\phi$

$\frac{\vec{A} \cdot \vec{B}}{A \cdot B} = \cos\phi = \frac{6}{\sqrt{22} \cdot \sqrt{17}} = \frac{6}{19,34} = 0,310$ /1

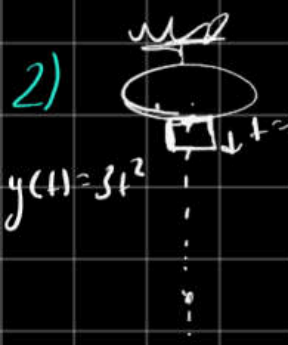
$\arccos\left(\frac{\vec{A}\vec{B}}{AB}\right) = \phi \qquad \cos(0,310) = 71,9407^\circ \cong 72^\circ$ /1

e) $\vec{R_3} = \vec{B} \times \vec{C} = ?$

$$\vec{R_3} = \vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -2 \\ 2 & 3 & 4 \end{vmatrix} = (8-(-6))\hat{i} - (12-(-4))\hat{j} + (9-4)\hat{k}$$

$$= 14\hat{i} - 16\hat{j} + 5\hat{k}$$

2)



a) Paket bırakıldığındaki hızı nedir?

$V_{i,paket} = V_{helikopter,\ t=2}$ → konum un türevi → hız denklemi

$y(t) = 6t = v(t)$

$v(2) = 12$ m/s

$V_{py} = 12$ m/s ↓

b) $t=2$ de helikopterin yüksekliği?

$y(2) = 3 \cdot 4 = 12$ m

c) Paket helikopterden bırakıldıktan kaç s sonra yere düşer?

$V_{ilk} = 12$ m/s
$h = 12$ m

$$y_s - y_i = h = V_0 t - \frac{1}{2} g t^2$$

$$-12 = +12 \cdot t - 5 \cdot t^2$$

$$5t^2 - 12t - 12 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{12 \pm \sqrt{144 - 4 \cdot 5 \cdot -12}}{10} = \frac{12 \pm 19{,}59}{10}$$

$3{,}16\,s$ ~~$t = -0{,}76$~~

d) Paketin yere ulaştığındaki hızı?

$y - y_0 = v_0 t - \frac{1}{2} g t^2$

$v_s = v_0 - g t$

$x = v_0 \cdot t$

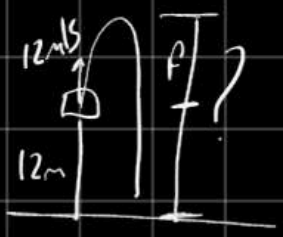
$v_s^2 = v_0^2 - 2g(y_s - y_i)$

$v_i = 12\,m/s$

$y - y_0 = -12\,m$

$v_s = 12 - 10 \cdot 3{,}16 = -19{,}6\,m/s$ ↓

e) Paketin ulaşabileceği maks mesafe?



$v_s = v_i - g t$

$0 = 12 - 10 \cdot t$

$t = 1{,}2\,s$

$f = y - y_0 = v_0 t - \frac{1}{2} g t^2$

$= 12 \cdot 1{,}2 - \frac{1}{2} \cdot 10 \cdot (1{,}2)^2$

$f = 7{,}2\,m$

$12 + 7{,}2 = 19{,}2\,m$ ↑

veya

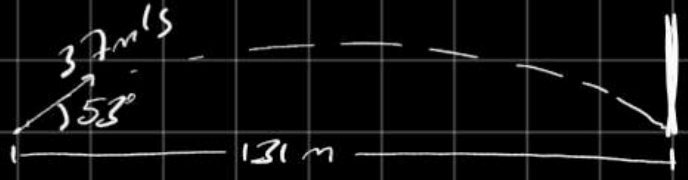
$h_{max} = \frac{v^2}{2g} = \frac{144}{20} = 7{,}2\,m$

$+12$

$= 19{,}2\,m$

3) Bir futbolcu 131 m uzaklıktaki kaleye doğru 53° açıyla ve 37 m/s hızla top vuruyor.

$g = 10 m/s$



a) $t = 2$ s de topun konumu, hızının büyüklüğü ve yönünü bulunuz.

$V_s = V_i - gt$ $\quad V_{iy} = 37 \cdot \sin 53 = 29,549 \simeq 29,55$ m/s

$V_{ix} = 37 \cdot \cos 53 = 22,267 \simeq 22,27$ m/s

$V_{s_y} = 29,55 - 10 \cdot 2 = 9,55$ m/s

$V_{s_x} = 22,27$ m/s

$V_s = \sqrt{(22,27)^2 + (9,55)^2} \simeq 24,23$ m/s

$\text{aradaki açı} = \arctan\left(\frac{9,55}{22,27}\right) \simeq 23,21°$

$x - x_0 = V_0 t + \frac{1}{2} a t^2$

$\Delta x = 22,2 \cdot t + \frac{1}{2} a t^2$

$\Delta x_{t=2} = 44,4$

$y - y_0 = V_0 t - \frac{1}{2} g t^2$

$y = 29,6 \cdot 2 - 20 = 39,2$

$y \sim 39$

$t = 2 \ (39,2 \text{ ile } 44,4 \text{ m})$

V_{ix} ~ V_{iy}

$V_{iy} = 29,6$ m/s

$y - y_0 = V_i t - \frac{1}{2} g t^2$

$0 = 29,6t - 5t^2$

$29,6t = 5t^2$

$t_{uç} = 5,92 = 6$ s

b) Topun havada en yüksek noktaya ulaştığı zamanı ve h yüksekliğini bulunuz.

$V_s = V_0 - gt = 0 \quad V_0 = g t_{ç} \quad t_{ç} = \frac{V_0}{g} = \frac{29,6}{10} = 2,965\,s$

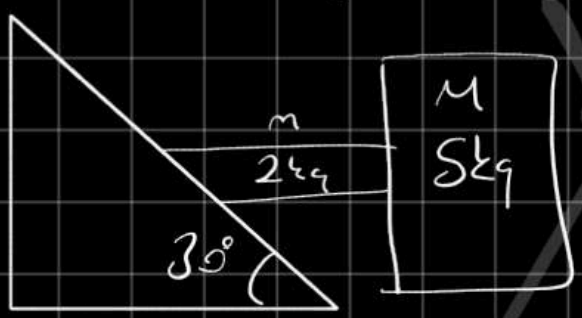
$y - y_0 = h = V_0 \cdot t - \frac{1}{2} g t^2 = 42,01\,m$

c) Topun menzili nedir? Kaleye ulaştı mı?

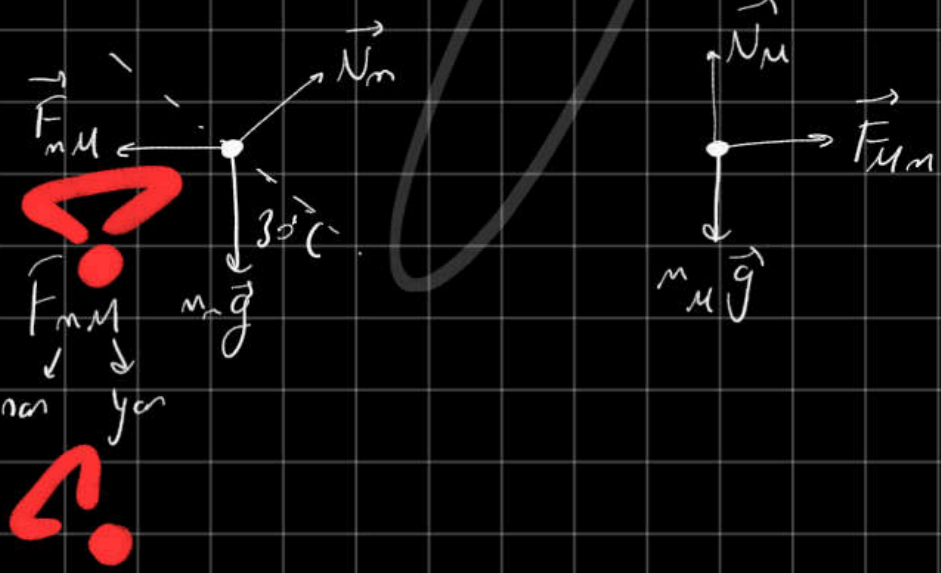
$R_{max} = \frac{V_0^2}{g} \sin 2\theta = \frac{(37)^2}{10} \cdot \sin 2 \cdot 53 = 131,59\,m$

$R = V_{0x} \cdot t = \checkmark$

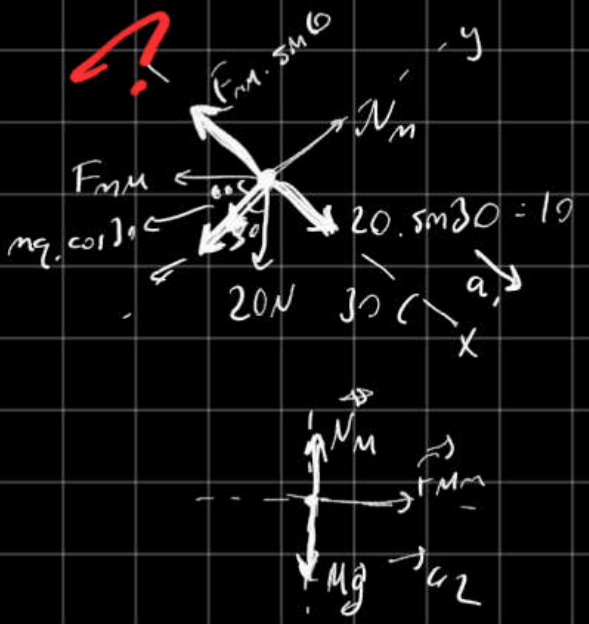
4)



a) Sistemin serbest cisim d. çizin



b) m ve M nin ivmeleri?



m için hareket denklemi:

$mg\sin 30 - F_{mM}\sin 60 = m.a_1$



$F_{Mm} = M.a_2 = M.a_1.\cos 30$

$mg\sin 30 - M.a_1.\cos 30.\sin 60 = m.a_1$

$m.a_1 + M.a_1.\cos 30.\sin 60 = mg\sin 30$

$a_1(m + M.\cos 30.\sin 60)$

$a_1 = \dfrac{mg\sin 30}{m + M.\cos 30.\sin 60}$

$= \dfrac{2.10.0,5}{2 + 5.(\cos 30)^2} = 1,74\ m/s^2$

$a_2 = 1,74.\cos 30 = 1,51\ m/s^2$

2023

1) $\vec{A} = -\hat{i} + 2\hat{j} - 2\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$ ve $\vec{C} = \hat{i} + 2\hat{j} + 3\hat{k}$

a) $\vec{R_1} = \vec{A} + 2\vec{B} - \vec{C} = ?$

$(-\hat{i} + 2\hat{j} - 2\hat{k}) + (4\hat{i} + 2\hat{j} - 2\hat{k}) + (-\hat{i} - 2\hat{j} - 3\hat{k})$

$= 2\hat{i} + 2\hat{j} - 7\hat{k}$ /

b) A, B ve C vekt. büyüklük? ?

$|\vec{A}| = \sqrt{(-1)^2 + 2^2 + (-2)^2} = \sqrt{9} = 3 = A$

$|\vec{B}| = \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{6} = 2,449 \cong 2,4 = B$

$|\vec{C}| = \sqrt{(-1)^2 + (-2)^2 + (-3)^2} = \sqrt{14} = 3,741 \cong 3,7 = C$ /

c) $R_2 = \vec{A} \cdot \vec{B} = ?$

$\vec{A} \cdot \vec{B} = A \cdot B \cdot \cos\phi = (A_x B_x)\hat{i}\hat{i} + (A_y B_y)\hat{j}\hat{j} + (A_z B_z)\hat{k}\hat{k}$

$= -2 \cdot 1 + 2 \cdot 1 + 2 \cdot 1 = 2$ /

d) A ve B vektörler arasındaki açı?

$\vec{A} \cdot \vec{B} = AB \cdot \cos\phi$

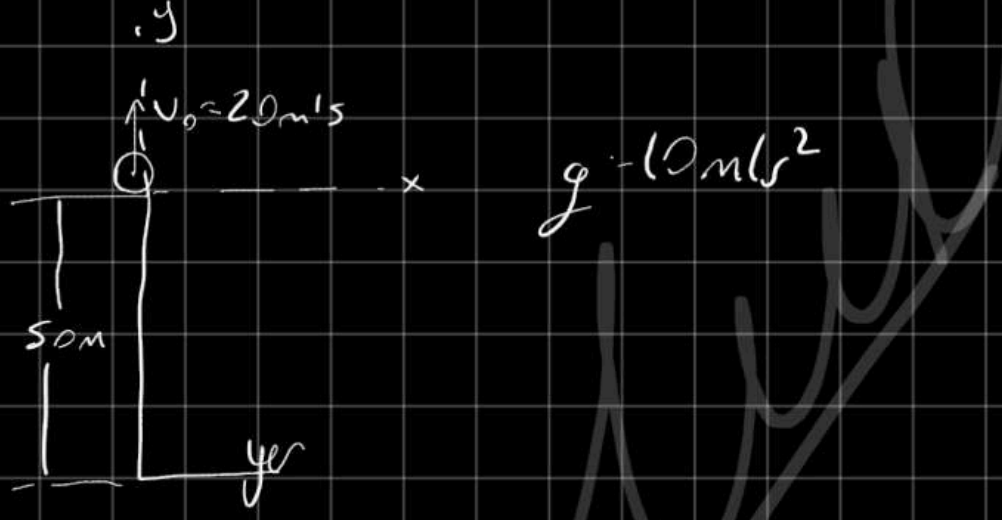
$\arccos\left(\frac{\vec{A} \cdot \vec{B}}{A \cdot B}\right) = \phi \quad \arccos\left(\frac{2}{3 \cdot 2,4}\right) = 73,872^\circ \cong 74^\circ$ /

e) $\vec{R_3} = \vec{B} \times \vec{C} = ?$

$$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 3 \end{vmatrix} = (3+2)\hat{i} - (6+1)\hat{j} + (4-1)\hat{k}$$

$$5\hat{i} - 7\hat{j} + 3\hat{k}$$

21)



a) Taş maksimum yüksekliğe ne kadar zamanda çıkar?

$V_0 = 20 m/s$, $h = 50m$, $g = 10 m/s^2$

$V_s = V_0 - gt \quad 0 = 20 - 10 \cdot t$

$t = 2 s$

b) Taşın erişeceği maks yükseklik?

$y - y_0 = V_0 t - \frac{1}{2} g t^2 \rightarrow h' = 20 \cdot 2 - \frac{1}{2} \cdot 10 \cdot 4$

$h' = 20m$

$h + h' = 20 + 50 = 70m$

c) Taş fırlatıldığı suyeye ne zaman gelir ve bu noktada hızı ne olur?

$V_s = V_0 - gt \quad -20 = 20 - gt$

$40 = 10 \cdot t$

$t = 4\,s$

hızı $= -20\,m/s$ olur

veya

$y - y_0 = 0 = V_0 t - \frac{1}{2} g t^2$

$0 = 20t - 5t^2$

$5t^2 = 20t \quad t = 4s \rightarrow V_s = V_0 - gt$

$V_s = 20 - 40 = -20\,m/s$

d) $t = 5s$ anında taşın hızı ve konumu nedir?

$V_s = V_0 - gt \rightarrow V_s = 20 - 10 \cdot 5$

$V_{t=5s} = -30\,m/s$

$y - y_0 = V_0 \cdot t - \frac{1}{2} g t^2$

$20 \cdot 5 - 5 \cdot 25$

$y - y_0 = -25$

$y = -25\,m$

e) Taş fırlatıldıktan kaç s sonra yere çarpar?

$$y - y_0 = V_0 t - \frac{1}{2} g t^2$$

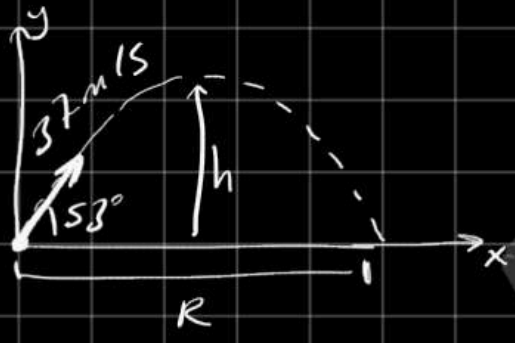
$$-50 = 20t - 5t^2$$

$$-10 = 4t - t^2$$

$$t^2 - 4t - 10 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{4 \pm \sqrt{56}}{2} = \frac{4 + 7,48}{2} = \underline{5,74}$$

3)



$g = 10 m/s^2$

$\sin 53 = 0,8$

$\cos 53 = 0,6$

a) $t = 2s$ de taşın konumu, hızının büyüklüğü ve yönü?

$V_0 = 37 m/s$

$V_{0y} = 37 \cdot \sin 53 = 29,54 \cong 29,6 \ m/s$

$V_{0x} = 37 \cdot \cos 53 = 22,26 \cong 22,3 \ m/s$

$V_s = V_0 - gt$ $\quad V_{sy} = 29,6 - 10 \cdot 2 = 9,6 m/s$

$V_{sx} = 22,3 m/s$

$\hookrightarrow V_{t=2} = \sqrt{(9,6)^2 + (22,3)^2} = 24,27 \cong \underline{24,3 \ m/s}$

$\arctan\left(\frac{V_y}{V_x}\right) = \phi \rightarrow \arctan\left(\frac{9,6}{22,3}\right) \cong \underline{23,3^\circ}$

$y - y_0 = V_0 t - \frac{1}{2} g t^2$

$y_{s_y} = 29,6 \cdot 2 - \frac{1}{2} \cdot 10 \cdot 4 = 39,2\ m$

$y_{s_x} = V_{0x} \cdot t \rightarrow y_{s_x} = 22,3 \cdot 2 = 44,6\ m$

$(39,2\ m, 44,6\ m)$

b) en yüksek noktaya ulaştığı zamanı ve h yüksekliği?

$V_s = V_0 - gt \rightarrow 0 = 29,6 - 10 \cdot t$

$t = 2,96\ s$

$y - y_0 = h = 29,6 \cdot 2,96 - \frac{1}{2} \cdot 10 \cdot 2,96 = 42,01\ m$

c) Yatay menzil R?

$y - y_0 = V_0 t - \frac{1}{2} g t^2 \quad 0 = 29,6 \cdot t - \frac{1}{2} 10 \cdot t^2$

$5t^2 = 29,6t$

$t_{uçuş} = 5,92\ s$

$x = V \cdot t \quad R = 22,2\ (5,92) = 131,42$

d) Topun izlediği yörüngenin y(x) denklemini yazın.

$V_{0x} = V_0 \cdot \cos\phi \ , \ V_{0y} = V_0 \cdot \sin\phi$

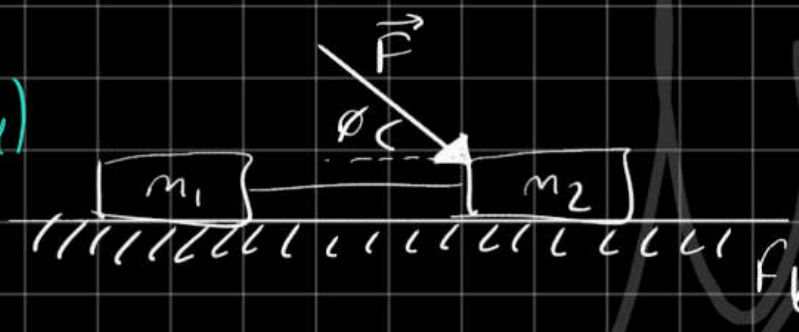
$x = V_0 t = V_0 \cdot \cos\phi \cdot t \rightarrow t = \frac{x}{V_0 \cdot \cos\phi}$

$$y = V_0 t - \frac{1}{2} g t^2 = V_0 \cdot \sin\phi \left(\frac{x}{V_0 \cdot \cos\phi}\right) - \frac{1}{2} \cdot g \cdot \left(\frac{x}{V_0 \cdot \cos\phi}\right)^2$$

$$= \frac{V_0 \cdot \sin\phi \cdot x}{V_0 \cdot \cos\phi} - \left(\frac{g \cdot x^2}{2 V_0^2 \cdot \cos^2\phi}\right)$$

$$= \tan\phi \cdot x - \frac{g}{2}\left(\frac{x}{V_0 \cdot \cos\phi}\right)^2$$

4)



$f_{bnetik} = \mu_k$

a) Her blok için serbest cisim diyagramı?



b) Her blok için Newton'un hareket denklemi ($\vec{F}_{net} = m \cdot \vec{a}$) yaz.

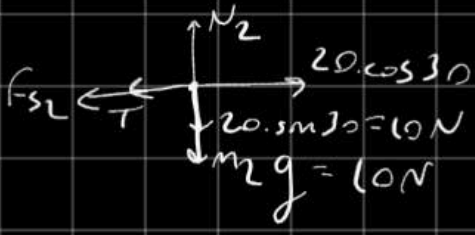
$m_1 \rightarrow T - f_k = m_1 \cdot a = T - \mu_k \cdot m_1 g$

$m_2 \rightarrow F\cos\phi - T - F_k = m_2 \cdot a$

$= F\cos\phi - T - \mu_k (F\sin\phi + m_2 g) = m_2 a$

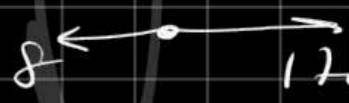


c) $m_1 = 2\,kg$, $m_2 = 1\,kg$, $\varphi = 30°$, $F = 20N$ ve $\mu_k = 0,2$ $a = ?$



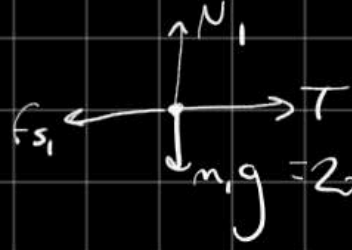
$N_2 = 10 + 20 = 20N$

$f_{s_2} = 0,2 \cdot 20 = 4N$



$9,4 = 3 \cdot a$

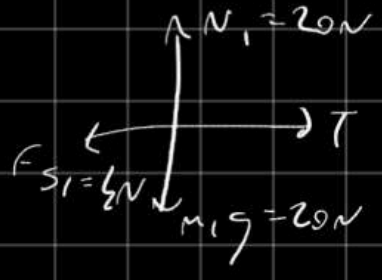
$a = 3,13\ m/s^2$



$N_1 = 20$

$f_{s_1} = 4N$

d) $T = ?$



$T - 4 = 2 \cdot 3,13$

$T = 10,26\ N$

